

Literature Status: Summable Szlenk Index and Injective Tensor Products

FA/Banach run `fa_banach.001`

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Status

This is a literature-already-answered packet, not an original proof from the run. Draga–Kochanek, arXiv:1604.03461, ask whether

$$X \widehat{\otimes}_{\varepsilon} Y$$

has summable Szlenk index whenever X and Y are separable Banach spaces with summable Szlenk index. Causey, arXiv:1707.08170, answers this affirmatively, and in a stronger form.

Original Question

The original source is:

S. Draga and T. Kochanek, *The Szlenk power type and tensor products of Banach spaces*, arXiv:1604.03461.

At the end of the paper, after proving results on Szlenk power type and separable determination, the authors ask whether the injective tensor product of two separable Banach spaces with summable Szlenk index must again have summable Szlenk index.

(Condition (ii) is satisfied, because the basis $(e_n^*)_{n=1}^\infty$ is also 1-unconditional.)

Choose any $q > \mathfrak{p}(E)$. By [6, Thm. 4.8] (see also [4, Prop. 2.3]), there exist a norm $|\cdot|$ on E and $c > 0$ with the following properties:

- (i) $\frac{1}{2}\|x\| \leq |x| \leq \|x\|$ for every $x \in E$;
- (ii) if $x^* \in E^*$ and $(x_n^*)_{n=1}^\infty$ is a weak*-null sequence with $|x_n^*| \geq \tau$ for some $\tau > 0$ and every $n \in \mathbb{N}$, then

$$\liminf_{n \rightarrow \infty} |x^* + x_n^*| \geq (|x^*|^q + c\tau^q)^{1/q}.$$

(We use the same symbol for the corresponding dual norm; of course, $\|x^*\| \leq |x^*| \leq 2\|x^*\|$ for every $x^* \in E^*$.)

Since F_n 's are finite-dimensional, we have $\lim_{\nu \rightarrow \infty} \|x_{\alpha \sim \nu, n}^*\| = 0$ for any $\alpha \in S_{N-1}$ and $n \in \mathbb{N}$. It means that each sequence of the form $(\sum_{n=1}^\infty \|x_{\alpha \sim \nu, n}^*\| e_n^*)_{\nu > \max \alpha}$, where $\alpha \in S_{N-1}$, is weak*-null. Therefore, using properties of the tree $(\mathbf{x}_\alpha^*)_{\alpha \in S_N}$ and the norm $|\cdot|$, we obtain

$$\begin{aligned} 1 &\geq \liminf_{\nu_2 \rightarrow \infty} \cdots \liminf_{\nu_N \rightarrow \infty} \left\| \sum_{j=1}^N \mathbf{x}_{(\nu_1, \dots, \nu_j)}^* \right\| = \liminf_{\nu_2 \rightarrow \infty} \cdots \liminf_{\nu_N \rightarrow \infty} \left\| \sum_{n=1}^\infty \left\| \sum_{j=1}^N x_{(\nu_1, \dots, \nu_j), n}^* \right\| e_n^* \right\| \\ &= \liminf_{\nu_2 \rightarrow \infty} \cdots \liminf_{\nu_N \rightarrow \infty} \left\| \sum_{n=1}^\infty \sum_{j=1}^N \|x_{(\nu_1, \dots, \nu_j), n}^*\| e_n^* \right\| \\ &\geq \frac{1}{2} \liminf_{\nu_2 \rightarrow \infty} \cdots \liminf_{\nu_N \rightarrow \infty} \left| \sum_{j=1}^N \sum_{n=1}^\infty \|x_{(\nu_1, \dots, \nu_j), n}^*\| e_n^* \right| \\ &\geq \frac{1}{2} \left(\left| \sum_{n=1}^\infty \|x_{(\nu_1), n}^*\| e_n^* \right|^q + c(N-1) \left(\frac{1}{8} \varepsilon \right)^q \right)^{1/q} \geq \frac{1}{16} (cN)^{1/q} \varepsilon. \end{aligned}$$

Thus, for some constant $C > 0$, we have $N \leq C\varepsilon^{-q}$ and hence $\mathfrak{p}(X) \leq q$. $\square$

We conclude the paper with posing a question closely related to our present work.

Question. Suppose X and Y are separable Banach spaces with summable Szlenk index. Does necessarily $X \hat{\otimes}_\varepsilon Y$ have summable Szlenk index as well?

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REFERENCES

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2. D. Alspach, R. Judd, E. Odell, *The Szlenk index and local ℓ_1 -indices*, Positivity **9** (2005), 1–44.
3. R. Causey, *Estimation of the Szlenk index of Banach spaces via Schreier spaces*, Studia Math. **216** (2013), 149–178.
4. S. Draga, T. Kochanek, *Direct sums and summability of the Szlenk index*, J. Funct. Anal. (2016), <http://dx.doi.org/10.1016/j.jfa.2016.02.020>

Later Answer

The supporting source is:

R. M. Causey, *Concerning summable Szlenk index*, arXiv:1707.08170.

Causey's abstract states that the paper proves that the injective tensor product of two Banach spaces has summable Szlenk index if both spaces do, and that this answers a question from Draga-Kochanek. The operator-level statement is Theorem 1.3: if A_0 and A_1 have summable Szlenk index, then so does

$$A_0 \otimes A_1 : X_0 \widehat{\otimes}_\varepsilon X_1 \longrightarrow Y_0 \widehat{\otimes}_\varepsilon Y_1,$$

with the converse when neither operator is zero.

Theorem 1.3. *Let $A_0 : X_0 \rightarrow Y_0$, $A_1 : X_1 \rightarrow Y_1$ be bounded, linear operators. If A_0, A_1 have summable Szlenk index, then so does the induced operator $A_0 \otimes A_1 : X_0 \hat{\otimes}_\varepsilon X_1 \rightarrow Y_0 \hat{\otimes}_\varepsilon Y_1$ between the injective tensor products. If neither A_0 nor A_1 is the zero operator, then the converse holds as well.*

One last application of Theorem 1.1 is a short proof of an operator version of a result from [8].

Theorem 1.4. *Let Λ be a non-empty set and let $A_\lambda : X_\lambda \rightarrow Y_\lambda$ be a uniformly bounded collection of linear operators. Then the induced operator $A : (\oplus_{\lambda \in \Lambda} X_\lambda)_{c_0(\Lambda)} \rightarrow (\oplus_{\lambda \in \Lambda} Y_\lambda)_{c_0(\Lambda)}$ has summable Szlenk index if and only if the operators A_λ have uniformly summable Szlenk index.*

We also study the ideal properties of the class $\mathfrak{S}$ of operators with summable Szlenk index, as well as introduce a way to assign to each operator A a value $\Sigma(A) \in [0, \infty]$ such that A has summable Szlenk index if and only if $\Sigma(A) < \infty$. Moreover, the quantity $\mathfrak{s}(A) := \|A\| + \Sigma(A)$ defines an ideal norm on $\mathfrak{S}$. In this direction, we prove the following.

Theorem 1.5. *The class $(\mathfrak{S}, \mathfrak{s})$ is a Banach ideal.*

We also study the behavior of summable Szlenk index of ℓ_p direct sums of operators for $1 \leq p \leq \infty$. Such a study is trivial in the setting of spaces, since the norm of the identity operator of a Banach space is either 0 or 1, but non-trivial for operators. We prove the following.

Theorem 1.6. *Let Λ be a non-empty set and let $A_\lambda : X_\lambda \rightarrow Y_\lambda$ be a uniformly bounded collection of linear operators. Then for any $1 \leq p \leq \infty$, the induced operator $A : (\oplus_{\lambda \in \Lambda} X_\lambda)_{\ell_p(\Lambda)} \rightarrow (\oplus_{\lambda \in \Lambda} Y_\lambda)_{\ell_p(\Lambda)}$ has summable Szlenk index if and only if $(\|A_\lambda\|)_{\lambda \in \Lambda} \in c_0(\Lambda)$ and $(\Sigma(A_\lambda))_{\lambda \in \Lambda} \in \ell_p(\Lambda)$.*

2. DEFINITIONS

Throughout, $\mathbb{K}$ will denote the scalar field (either $\mathbb{R}$ or $\mathbb{C}$), and “operator” will mean “bounded, linear operator.”

Given a directed set D and $n \in \mathbb{N}$, we let $D^{\leq n} = \cup_{i=1}^n D^i$. Given $t = (u_i)_{i=1}^k \in D^{\leq n}$, we let $|t| = k$, $t^- = (u_i)_{i=1}^{k-1}$ (where $(u_i)_{i=1}^0 = \emptyset$ by convention), and for $0 \leq j \leq k$, $t|_j = (u_i)_{i=1}^j$. Given $t = (u_i)_{i=1}^k \in \{\emptyset\} \cup D^{\leq n-1}$ and $u \in D$, we let $t \frown u = (u_1, \dots, u_k, u) \in D^{\leq n}$. For $s, t \in D^{\leq n}$, we let $s \frown t$ be the concatenation of s and t . If X is a Banach space, we say a collection $(x_t)_{t \in D^{\leq n}} \subset X$ is *weakly null* provided that for every $t \in \{\emptyset\} \cup D^{\leq n-1}$, $(x_{t \frown u})_{u \in D}$ is a weakly null net. We say a map $\phi : D^{\leq n} \rightarrow D^{\leq n}$ is a *pruning* provided that $|\phi(t)| = |t|$ and $\phi(t)^- = \phi(t^-)$ for each $t \in D^{\leq n}$ and such that the collection $(x'_t)_{t \in D^{\leq n}}$ is weakly null, where $x'_t = x_{\phi(t)}$. The following can be easily proved by induction on n . We will use this result frequently.

Proposition 2.1. *Let D be a directed set, $n \in \mathbb{N}$, X a Banach space, and $(x_t)_{t \in D^{\leq n}}$ a weakly null collection. Let (M, d) be a compact metric space and suppose $F : D^n \rightarrow M$ is any function. Then for any $\delta > 0$, there exist a pruning $\phi : D^{\leq n} \rightarrow D^{\leq n}$ and $\varpi \in M$ such that $d(\varpi, \phi(t)) \leq \delta$ for any $t \in D^n$.*

For a Banach space X and $n \in \mathbb{N}$, we let $\{X\}_n$ denote the set of all norms $|\cdot|$ on $\mathbb{K}^n$ such that for any $b > 1$, there exists a weakly null collection $(x_t)_{t \in D^{\leq n}} \subset S_X$ such that for any $(a_i)_{i=1}^n \in S_{\ell_\infty^n}$

The space-level answer is Corollary 5.3:

$$X_0 \hat{\otimes}_\varepsilon X_1 \text{ has summable Szlenk index} \iff X_0 \text{ and } X_1 \text{ do,}$$

for non-zero Banach spaces X_0, X_1 . The same corollary gives the equivalent asymptotic c_0 formulation.

The following answers a question from [9].

Corollary 5.3. *Let X_0, X_1 be non-zero Banach spaces. Then $X_0 \hat{\otimes}_\varepsilon X_1$ is Asymptotic c_0 if and only if X_0, X_1 are. Equivalently, $X_0 \hat{\otimes}_\varepsilon X_1$ has summable Szlenk index if and only if X_0, X_1 do.*

6. DIRECT SUMS

The first result of this section is an operator version of a result from [8]. However, we will use Corollary 3.4 to give a new proof.

Theorem 6.1. *Suppose that Λ is a non-empty set and for each $\lambda \in \Lambda$, $A_\lambda : X_\lambda \rightarrow Y_\lambda$ is an operator. Assume also that $\sup_{\lambda \in \Lambda} \|A_\lambda\| < \infty$ and let $A : (\oplus_{\lambda \in \Lambda} X_\lambda)_{c_0(\Lambda)} \rightarrow (\oplus_{\lambda \in \Lambda} Y_\lambda)_{c_0(\Lambda)}$ be the operator such that $A|_{X_\lambda} = A_\lambda$.*

Then A has summable Szlenk index if and only if there exists M such that for each $\lambda \in \Lambda$, A_λ has M -summable Szlenk index.

Proof. It is clear that if A has M -summable Szlenk index, A_λ has M -summable Szlenk index for each $\lambda \in \Lambda$, which gives one direction.

Now suppose there exists M such that A_λ has M -summable Szlenk index for each $\lambda \in \Lambda$. Let $K = \bigcup_{\lambda \in \Lambda} A_\lambda^* B_{Y_\lambda^*}$. It is clear that for any $n \in \mathbb{N}$ and $\varepsilon_1, \dots, \varepsilon_n$,

$$s_{\varepsilon_1} \dots s_{\varepsilon_n}(K) \subset \{0\} \cup \bigcup_{\lambda \in \Lambda} s_{\varepsilon_1} \dots s_{\varepsilon_n}(A_\lambda^* B_{Y_\lambda^*}).$$

From this it follows that with $M' = M + 2 \sup_{\lambda \in \Lambda} \|A_\lambda\|$, K has M' -summable Szlenk index. Indeed, suppose $\varepsilon_1, \dots, \varepsilon_n > 0$ are such that $\sum_{i=1}^n \varepsilon_i > M'$. Note that $\varepsilon_i \leq 2 \sup_{\lambda \in \Lambda} \|A_\lambda\|$ for each $1 \leq i \leq n$. This means that $\sum_{i=2}^n \varepsilon_i > M$, whence

$$s_{\varepsilon_1} \dots s_{\varepsilon_n}(K) \subset s_{\varepsilon_1} \left(\{0\} \cup \bigcup_{\lambda \in \Lambda} s_{\varepsilon_2} \dots s_{\varepsilon_n}(A_\lambda^* B_{Y_\lambda^*}) \right) = s_{\varepsilon_1}(\{0\}) = \emptyset.$$

□

We next turn to a facet of this problem which is of interest for operators, but not for spaces. Above we considered c_0 direct sums, while below we wish to consider ℓ_p direct sums, $1 \leq p \leq \infty$. However, if $(X_\lambda)_{\lambda \in \Lambda}$ is a collection of non-zero Banach spaces, $(\oplus_{\lambda \in \Lambda} X_\lambda)_{\ell_p(\Lambda)}$ contains a copy of ℓ_p and therefore cannot have summable Szlenk index except in the case that Λ is finite. Our final goal is to elucidate the situation for operators.

Proposition 6.2. *Fix $1 \leq p \leq \infty$. For any operators $A_i : X_i \rightarrow Y_i$, $1 \leq i \leq k$ and $n \in \mathbb{N}$,*

$$\Sigma_n \left(A : (\oplus_{i=1}^k X_i)_{\ell_p^k} \rightarrow (\oplus_{i=1}^k Y_i)_{\ell_p^k} \right) \leq \|(\Sigma_n(A_i))_{i=1}^k\|_{\ell_p^k}$$

and

$$\Sigma \left(A : (\oplus_{i=1}^k X_i)_{\ell_p^k} \rightarrow (\oplus_{i=1}^k Y_i)_{\ell_p^k} \right) = \|(\Sigma(A_i))_{i=1}^k\|_{\ell_p^k}.$$

Proof. In the proof, we identify X_i and Y_i with subspaces of $X = (\oplus_{i=1}^k X_i)_{\ell_p^k}$ and $Y = (\oplus_{i=1}^k Y_i)_{\ell_p^k}$, respectively. Let $A : X \rightarrow Y$ denote the operator with $A|_{X_j} = A_j$. Let $P_j : X \rightarrow X$ denote the projection from X onto X_j . Then $\Sigma_n(A_i) = \Sigma_n(AP_i)$.

Verification Notes

This is a separate-source answer. The original paper poses the question at the end; Causey’s later paper cites the Draga–Kochanek paper as reference [9] and explicitly says that the tensor-product result answers a question from that reference. Causey’s Corollary 5.3 is stronger than the original question because it removes separability and proves an if-and-only-if statement for non-zero spaces.

Search Bounds

The run registry, solution README, and active-claim directory were checked for overlapping entries involving arXiv:1604.03461, arXiv:1707.08170, Draga, Kochanek, Causey, summable Szlenk index, and injective tensor products. The original and supporting arXiv PDFs were downloaded and inspected locally. No MathSciNet or Zentralblatt search was performed.

Review Recommendation

Verify the final question in arXiv:1604.03461 and Corollary 5.3 in arXiv:1707.08170. The status transfer relies only on this explicit question and Causey’s explicit later answer.